

Krishisetu

Agriculture, FoodTech & Rural
Development

Smart Crop Advisory
System for Small and Marginal Farmers

Let's know more

Welcome to KrishiSetu

Your Agricultural Companion

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Krishisetu



25°C

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Clear

Farm Management & Planning JUL 17



Crop
Calender



Expert Advice



My Soil
Report

Market & Resources \$



Mandi Prices



Govt
Schemes



Tools & Tech

Knowledge & Learning 💡



Resource Hub



Agri-Learn



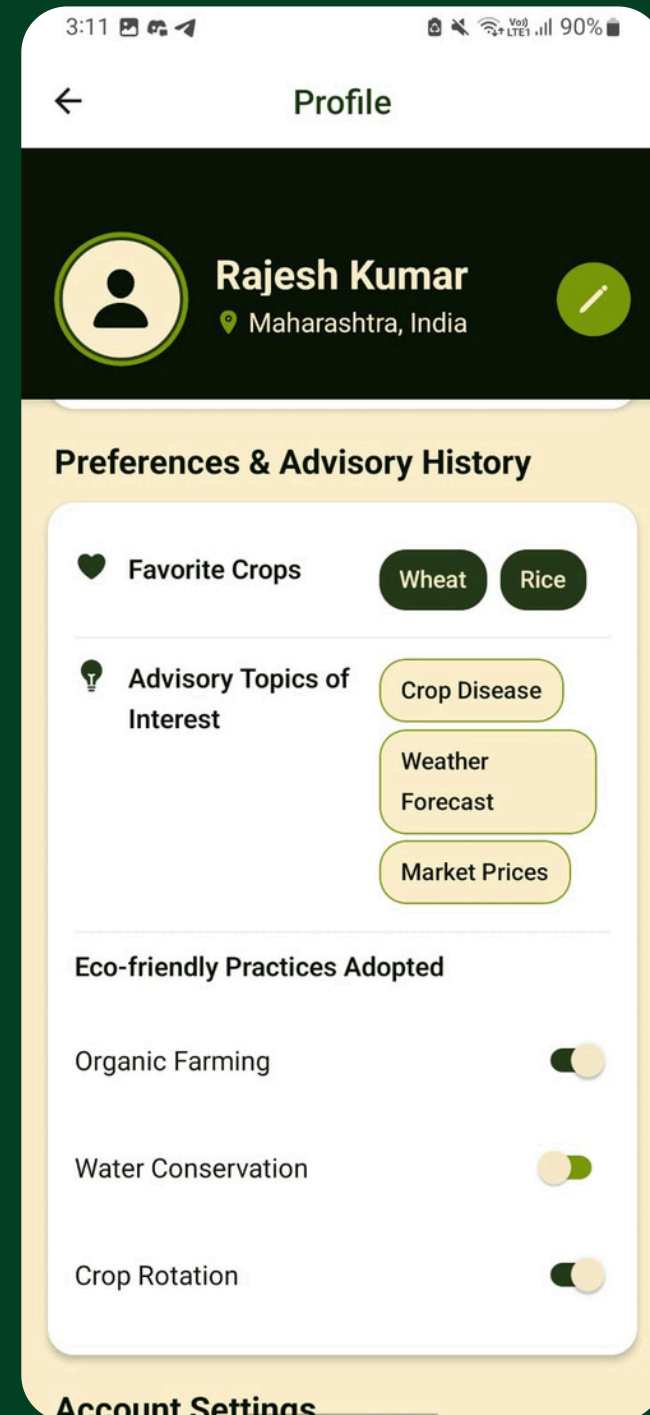
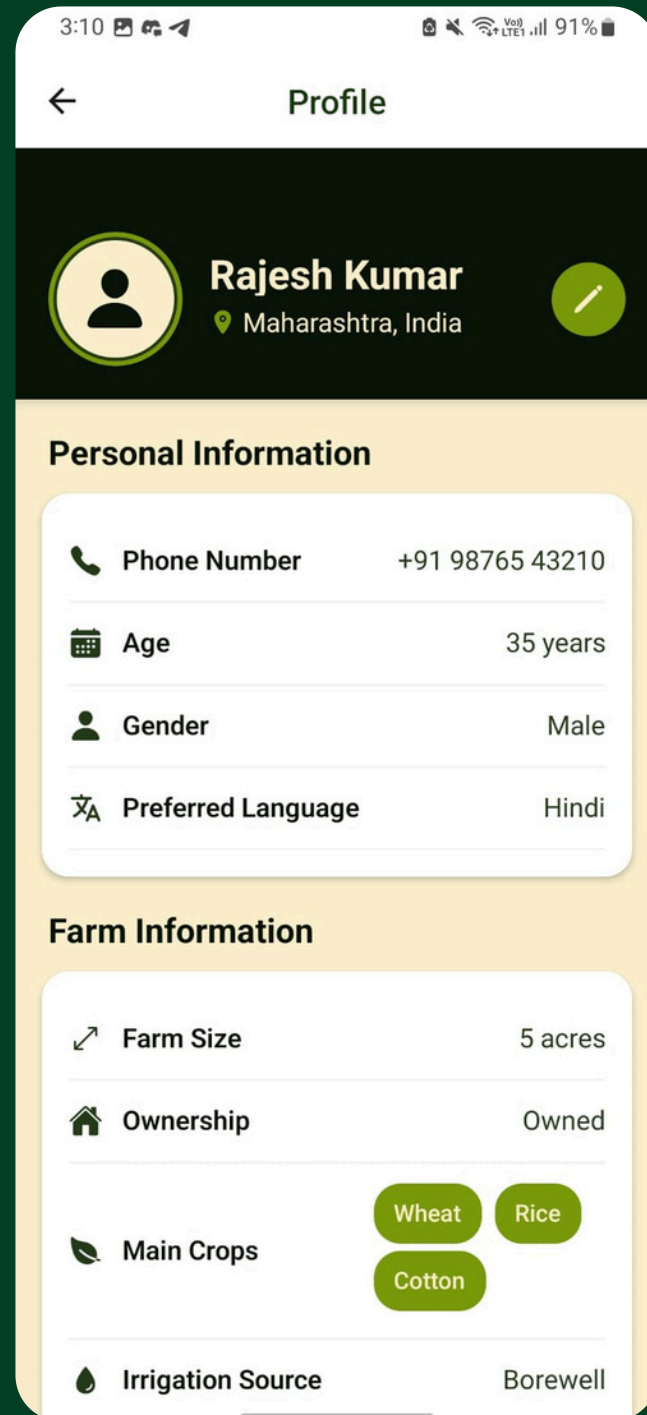
Tips & Tricks



Krishi Mitra

My Voice Assistant





About Krishisetu

“Imagine standing in your field, wondering: Should I water today? Which seed will thrive? What if tomorrow brings heavy rain?”

KrishiSetu isn't just an app—it's a digital farming companion. Multilingual, voice-first, always-on, and hyper-local.

👉 A farmer asks in her own dialect: “What crop should I sow this season?” → Krishisetu replies with soil-tested, weather-aligned, market-aware advice.

👉 She uploads a photo of a diseased leaf → AI detects the pest in seconds, prescribes eco-friendly treatment.

👉 A sudden rainfall alert arrives → She adjusts irrigation, saving water and money.

👉 Her crop calendar reminds her when to sow, irrigate, and harvest—synced with local weather.

👉 At harvest, Krishisetu shows nearest storage centers and best mandi prices → reducing losses, boosting income.

...Krishisetu

○ The Core Benefits (3 pillars)

Confidence – No more uncertainty. Every action is backed by data.

Simplicity – Talk, tap, or snap. Advice comes in plain language, instantly.

Empowerment – From soil to market, farmers control their journey.

○ What's Different:

Unlike static advisories, KrishiSetu learns and evolves with each farm, each plot, each season. Built for rural realities—voice-first, multilingual, offline-ready.

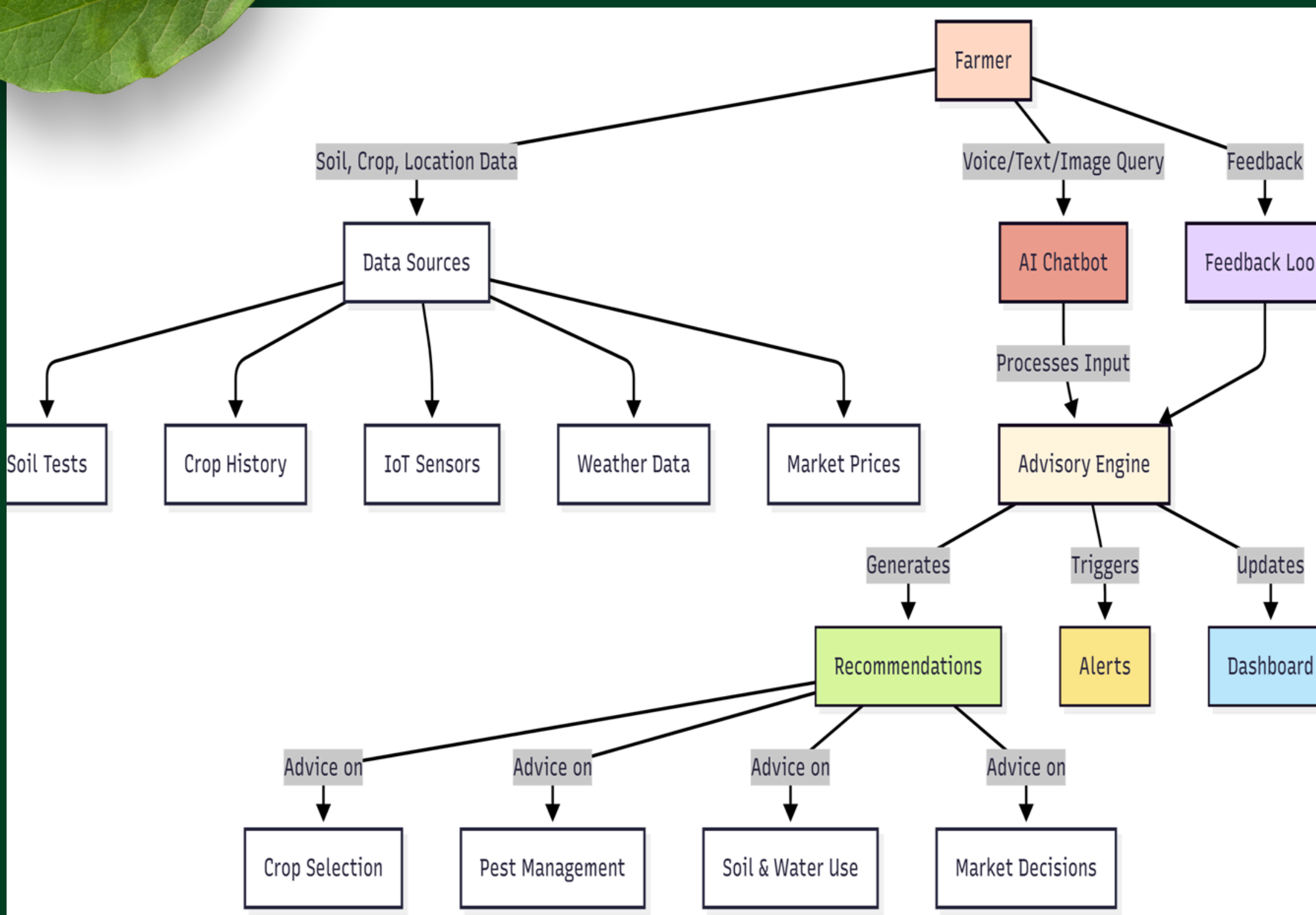
On site research...



Technical Approaches

Flow chart

Tech Stack



■ Frontend (Mobile App)

- Built with: React Native (Expo SDK) + TypeScript
- Navigation: Expo Router
- UI: Core RN Components, Ionicons, Custom Themes
- Libraries: expo-av, expo-speech, image/document picker, secure store, reanimated

💻 Backend (Server)

- Stack: Node.js + Express.js
- Security: JWT, bcrypt, CORS
- Env & Files: dotenv, Multer, ffmpeg-static, fluent-ffmpeg

🤖 Python Backend (AI & ML)

- Frameworks: Flask, Flask-CORS
- ML Tools: TensorFlow, Hugging Face
- Models: Crop & pest detection, image classification
- Image Processing: Pillow, OpenCV
- AI Integration: Google Gemini AI (chat, NLP, TTS, image analysis)
- Storage: In-memory (Dev), expo-file-system → MongoDB (Planned)

FEASIBILITY AND VIABILITY

FEASIBILITY & VIABILITY

1. Technical: Built on proven AI/ML + scalable cloud.
2. Operational: Farmers already use WhatsApp/voice apps → easy adoption.
3. Economic: Affordable via low-cost smartphones & govt. schemes.

CHALLENGES & RISKS

- i. Digital literacy gaps
- ii. Weak rural connectivity
- iii. AI errors may reduce trust
- iv. Data privacy concerns

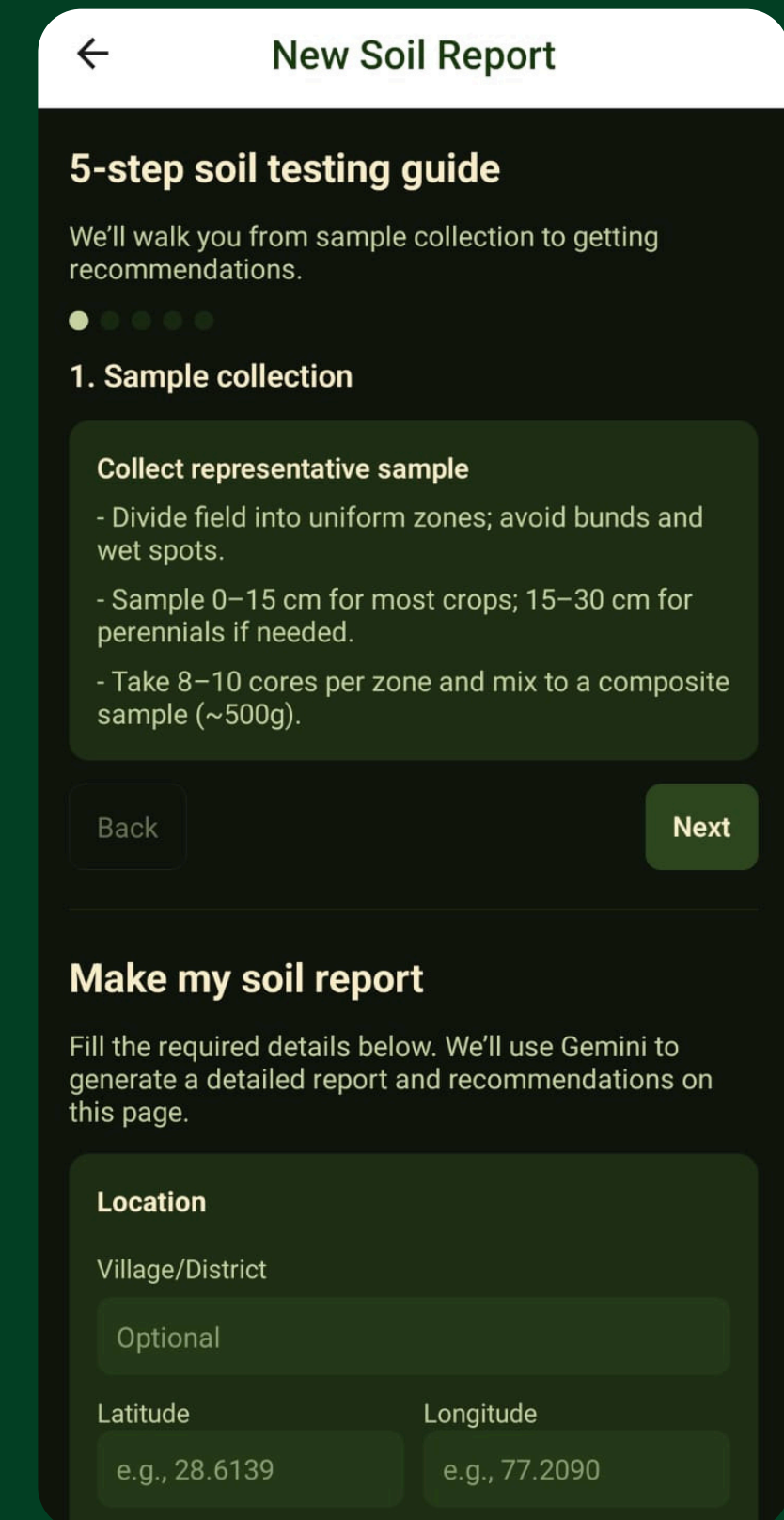
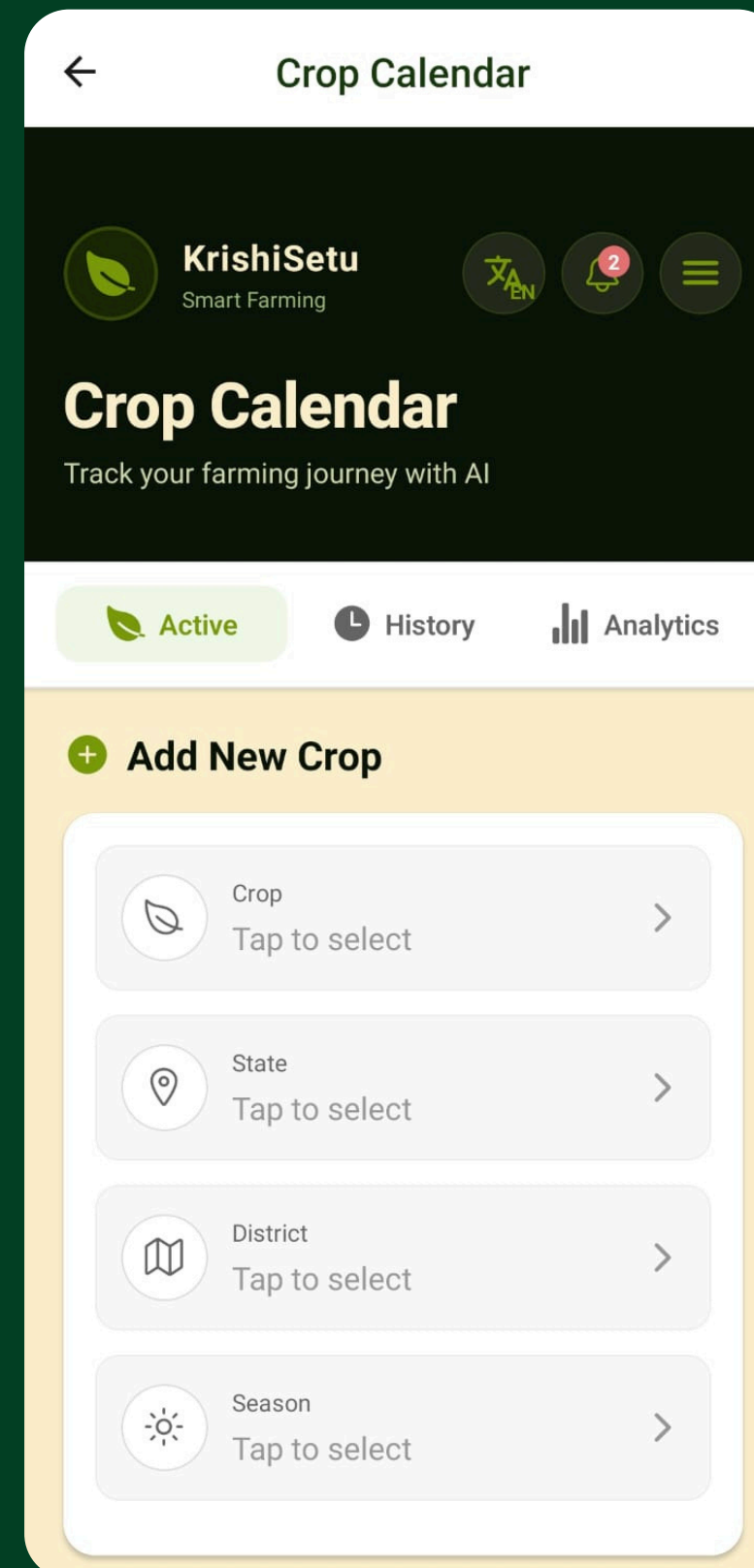
SMART STRATEGIES

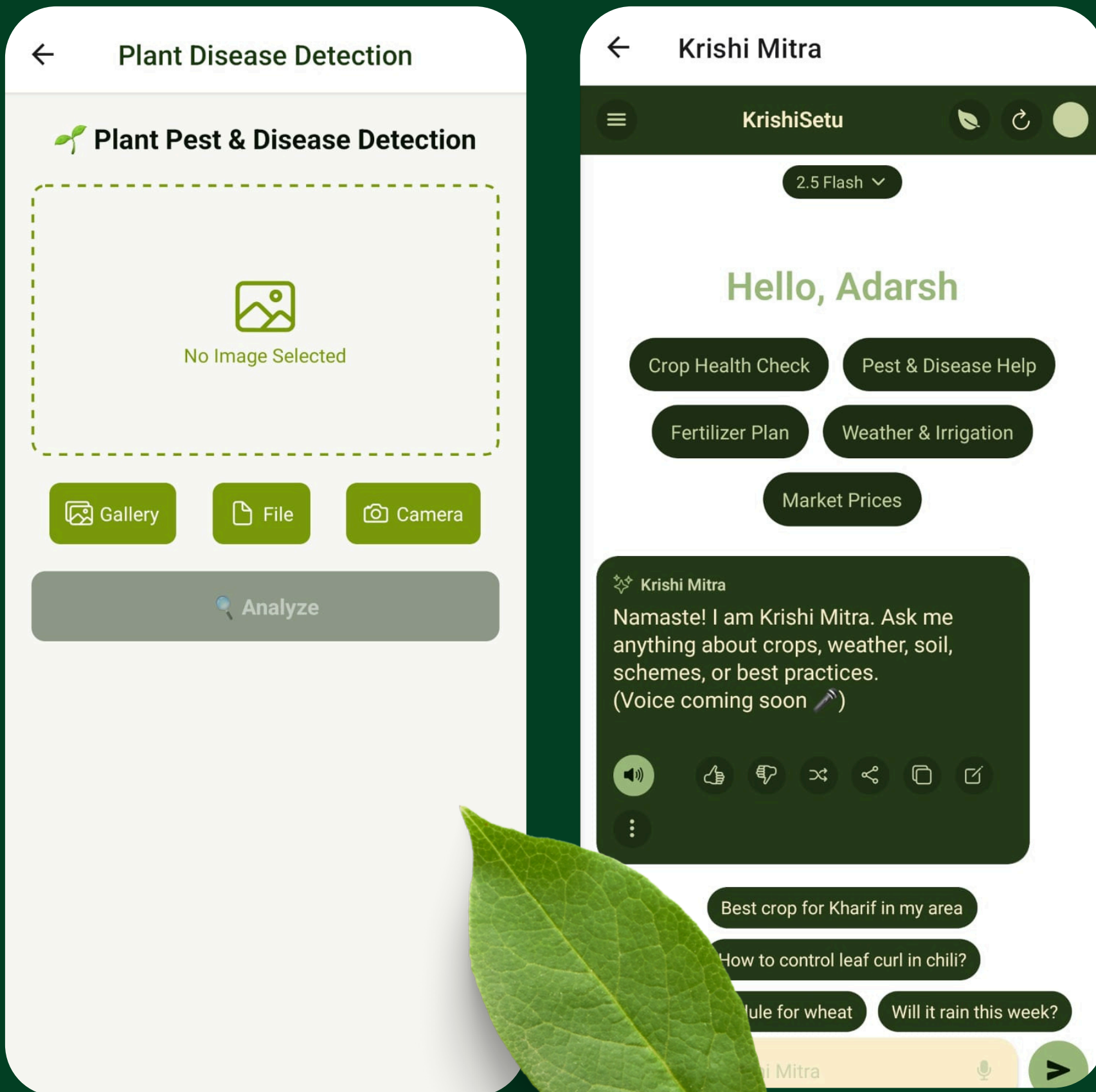
- a) Voice-first + local languages → break literacy barrier
- b) Offline & lightweight → works even with poor internet
- c) Expert validation → ensures AI accuracy
- d) Secure data policies → protect farmer trust

Impacts...

Potential Impact on Target Audience

- Empowers small & marginal farmers with real-time, personalized decisions.
- Reduces dependence on middlemen & unreliable advice.
- Builds farmer confidence → stronger adoption of modern agri-tech.





Benefits...

Social

- **Multilingual + voice-first → accessible for all farmers.**
- **Builds stronger, informed rural communities.**

Economic

- **20–30% higher yields.**
- **Lower costs via smart fertilizer & water use.**
- **Better mandi prices → fair income.**

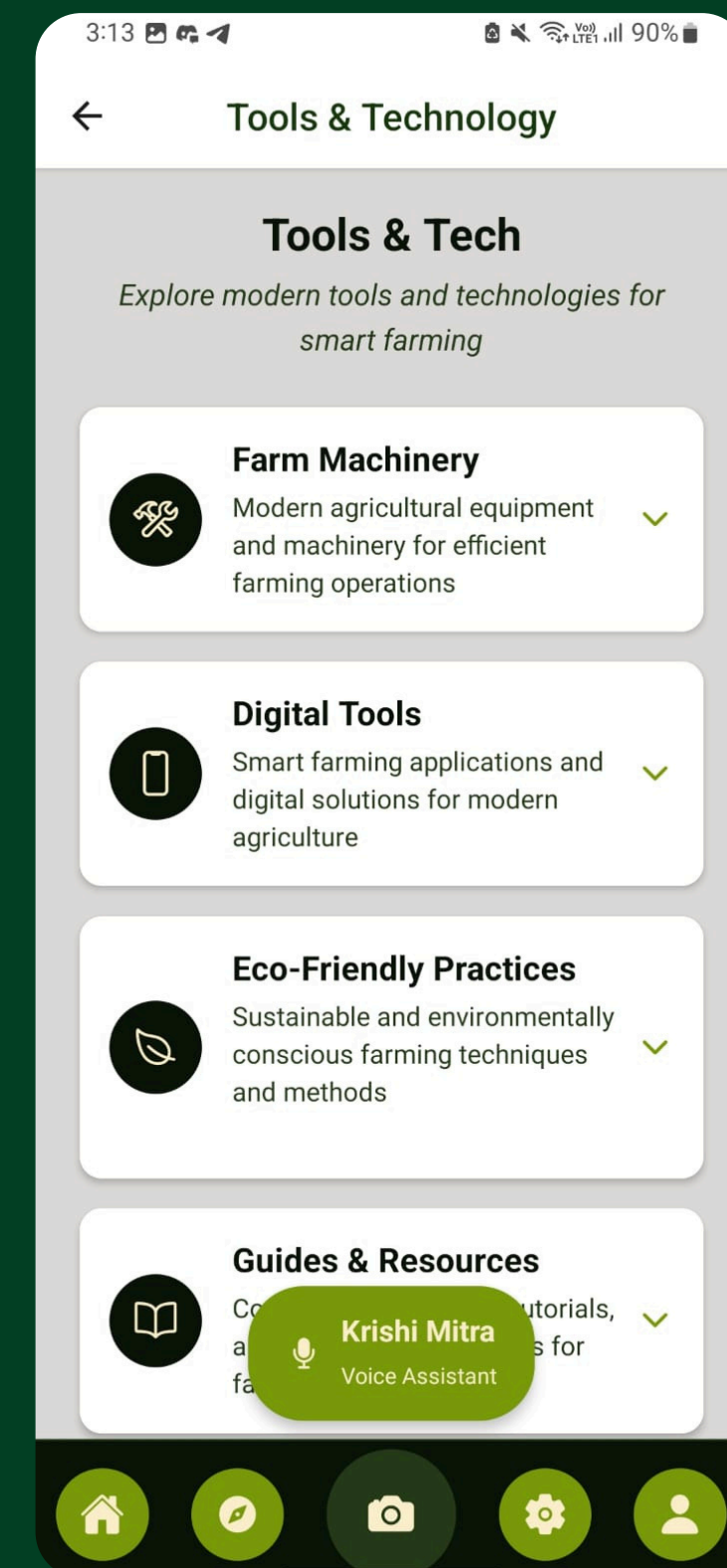
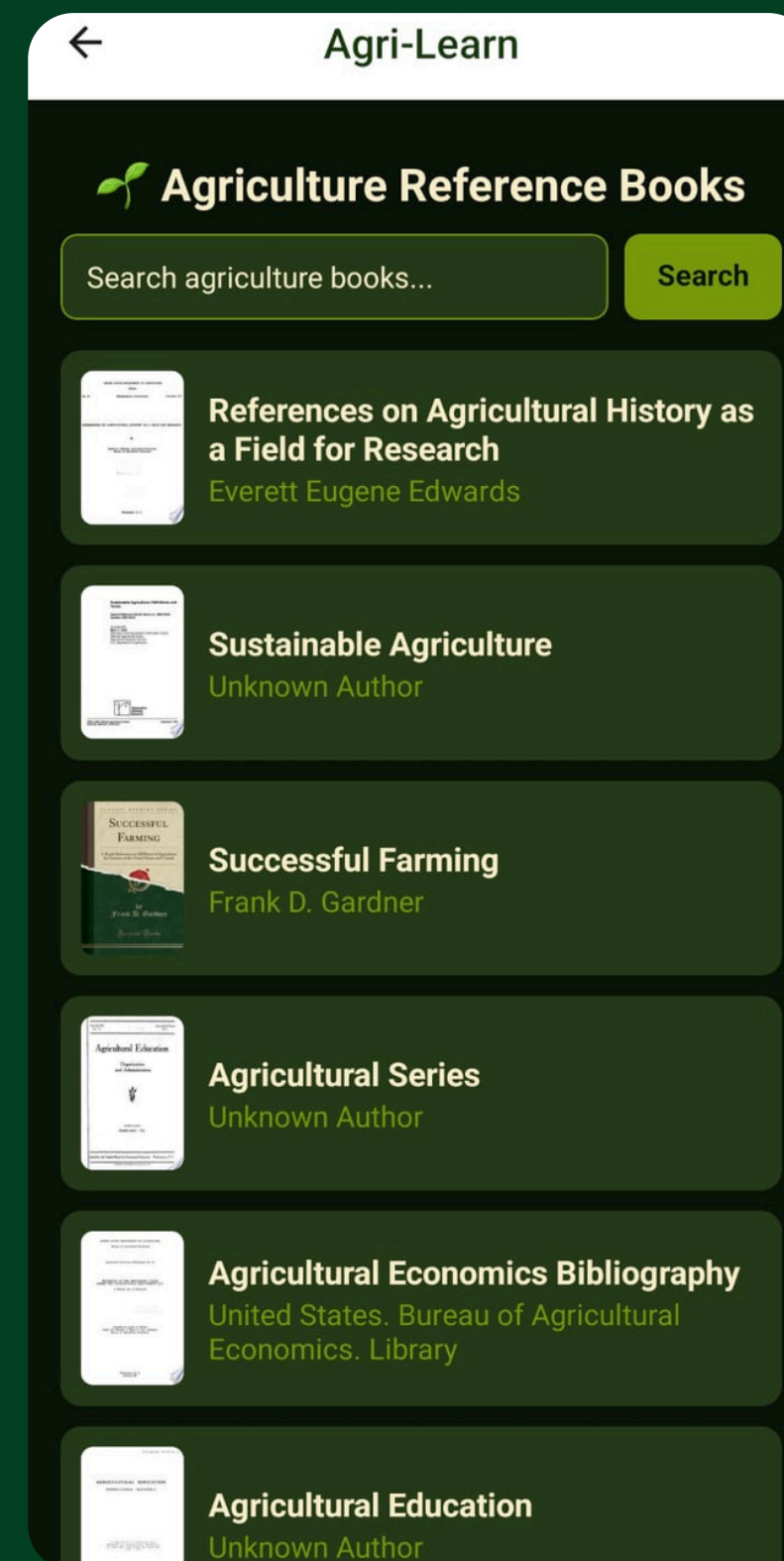
Environmental

- **Less chemical use → healthier soil & crops.**
- **Smart irrigation → saves water.**
- **Promotes sustainable farming.**

Research and References

Data Sources:

- Weather & Agro Data → Open-matro, meteorological department of India.
- Soil Health Data → Soil Health Card Scheme, SoilGrids, field lab reports.
- Crop & Pest/Disease Data → ICAR, FAO, PlantVillage dataset, annotated farmer images.
- Market Price Data → Agmarknet, e-NAM, mandi boards.
- Govt Schemes & Advisory → PM-KISAN, PMFBY, KCC, NABARD reports.
- Farmer Productivity Challenges → NABARD, “Low Productivity of Small & Marginal Farmers in India” report





Insights from research..

- a) 86% of farmers are small/marginal, with fragmented land & low tech access.
- b) Major gaps: real-time weather, pest alerts, storage, market awareness.
- c) ICT-based advisory can increase yields by 20–30%.

Thank You!

for listening us and hope you
understood the concept :)



www.krishisetu.com

